SML ASSIGNMENT 1

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Performance Summary

Model Comparison

The **Quadratic Discriminant Analysis (QDA)** model significantly outperformed the **Linear Discriminant Analysis (LDA)** model on this subset of the MNIST dataset (classes 0, 1, and 2).

- **LDA Accuracy**: ~88%
- **QDA Accuracy**: ~98%

The superior performance of QDA suggests that the classes are not well-represented by a single pooled covariance matrix. Each digit has a distinct shape and variance structure; for instance, the way pixels vary in a "1" is quite different from a "0" or a "2". By allowing for class-specific covariance matrices, QDA captures these individual distributions more accurately, leading to much higher classification precision.

t-SNE Visualization Interpretation

Looking at the plots, the three colors (digits 0, 1, and 2) are in very clear groups.

Digit 1 is very far away from the others, which makes sense because it's just a straight line.

Digit 0 and 2 are closer but still have their own space. Since the groups are so clear in 2D, it explains why the math models (especially QDA) could reach almost 100% accuracy.

The high degree of separation in the 2D t-SNE projection aligns with the high accuracy achieved by the QDA model. While LDA achieves decent results, its assumption of equal covariance likely forces a linear decision boundary that misclassifies samples where class distributions overlap or have significantly different spreads in the high-dimensional space.

Discriminant Value Example for the very first test sample (which is a 0):

LDA Scores: [162283,38531,64125] (Predicted: 0)

QDA Scores: [-2553360, -37689721, -8810431] (Predicted: 0)

Even though the numbers are different, both models gave class 0 the highest score, so they got it right!

Discriminant Scores: Analysis of individual test samples showed that QDA provided much stronger differentiation (higher confidence) in its discriminant scores than LDA.